Climate inequality

Matthew Alampay Davis*

Job market paper

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Abstract

This paper investigates the relationship between anthropogenic climate change and economic inequality, a subject intersecting two of the defining challenges of the 21st century but which remains critically understudied. To address this research gap, I first adapt tools developed in recent macroeconometric literature to overcome well-documented limitations of standard methods used in climate impact studies. In an expository application, I demonstrate features of this approach which enable credible, robust, and directly testable characterization of long-term dynamics which otherwise produce unstable estimates under prevailing approaches. I apply these refinements to newly available distributional national accounts data to document new evidence that temperature shocks exacerbate income inequality within countries, an effect driven by disproportionate reductions of income among the poorest segments of already-poor countries. To better understand the implications of this incidence pattern for the welfare economics of climate change, I integrate these results over observed income distributions and use output from cutting-edge climate models to simulate the contributions of anthropogenic radiative forcings on these climate inequalities. These findings constitute the first comprehensive characterization of the regressivity of past and future climate change.

*wm.alampaydavis@gmail.com

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